

Project Title: AI Usage and Academic Performance: A Data-Driven Analysis

Proposal

This project will analyze the relationship between AI tool usage and academic performance. The dataset will primarily be self-collected, focusing on daily AI usage patterns and study behaviors over a period of approximately 3–4 weeks. Data will be gathered by recording variables such as time spent using AI tools (e.g., ChatGPT, Gemini), purpose of use (learning, assignment help, quick answers), number of interactions, and duration of study sessions. In addition, performance-related variables such as assignment grades and self-reported productivity levels will be included. The main research question is whether AI usage has a positive or negative impact on academic performance and productivity.

Data collection will be conducted manually using a structured logging system (e.g., spreadsheet), where entries will be recorded daily. The dataset will include clearly defined features such as AI usage time, purpose of use, number of interactions, study duration, productivity level, and assignment performance. In addition, contextual features such as time of day, proximity to deadlines or exams, and type of academic task (e.g., coding, reading, writing) will be recorded. To improve the analysis, course-related variables will also be included, such as course type (quantitative vs. qualitative), perceived difficulty level, and whether the course involves project-based work.

The dataset is expected to consist of approximately 40–60 observations, each including multiple numerical and categorical features. The data will be structured and suitable for statistical analysis and machine learning applications. The project will apply exploratory data analysis, hypothesis testing, and predictive modeling techniques to examine whether AI usage has a measurable impact on academic performance and productivity across different types of courses.